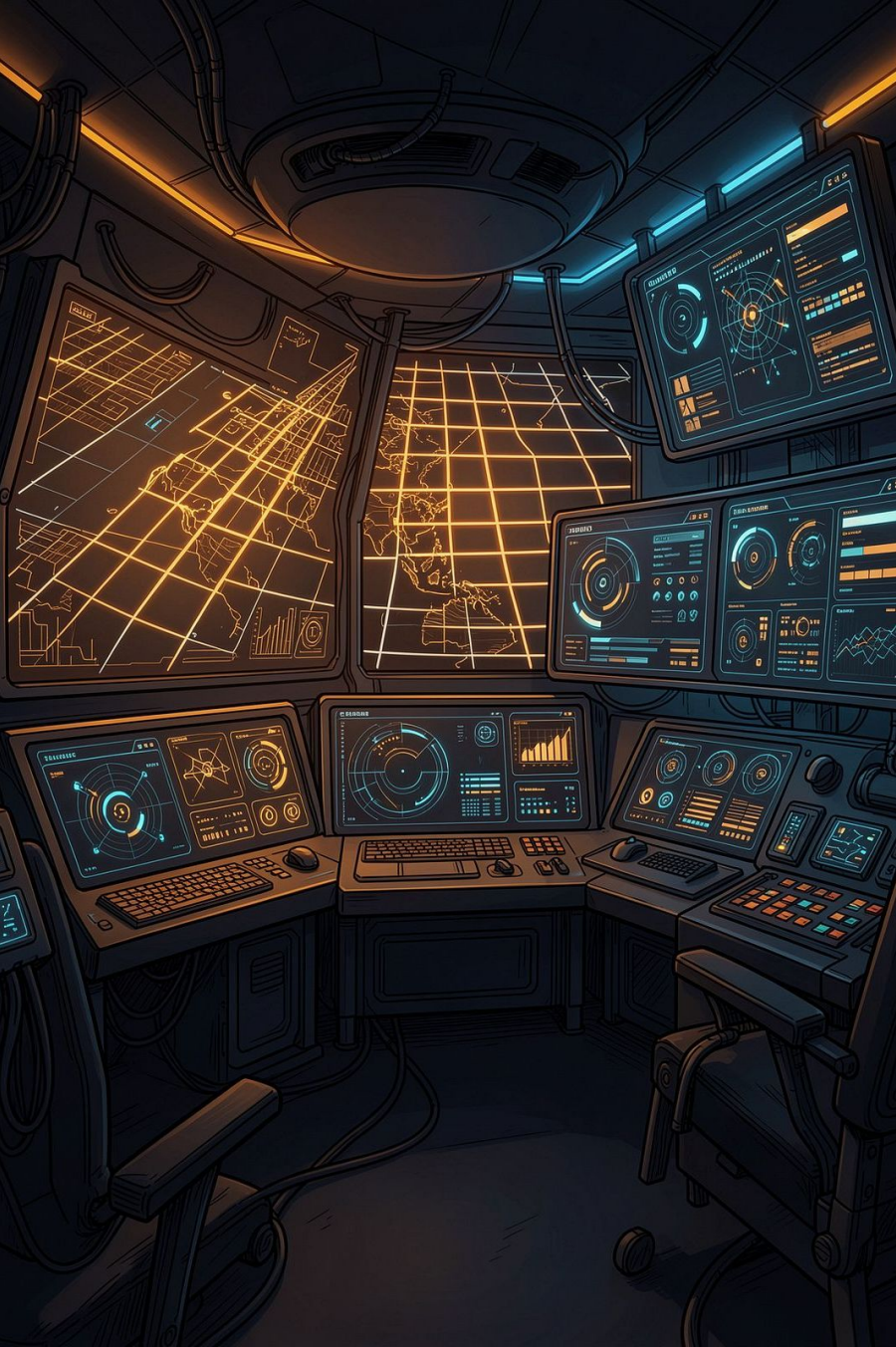
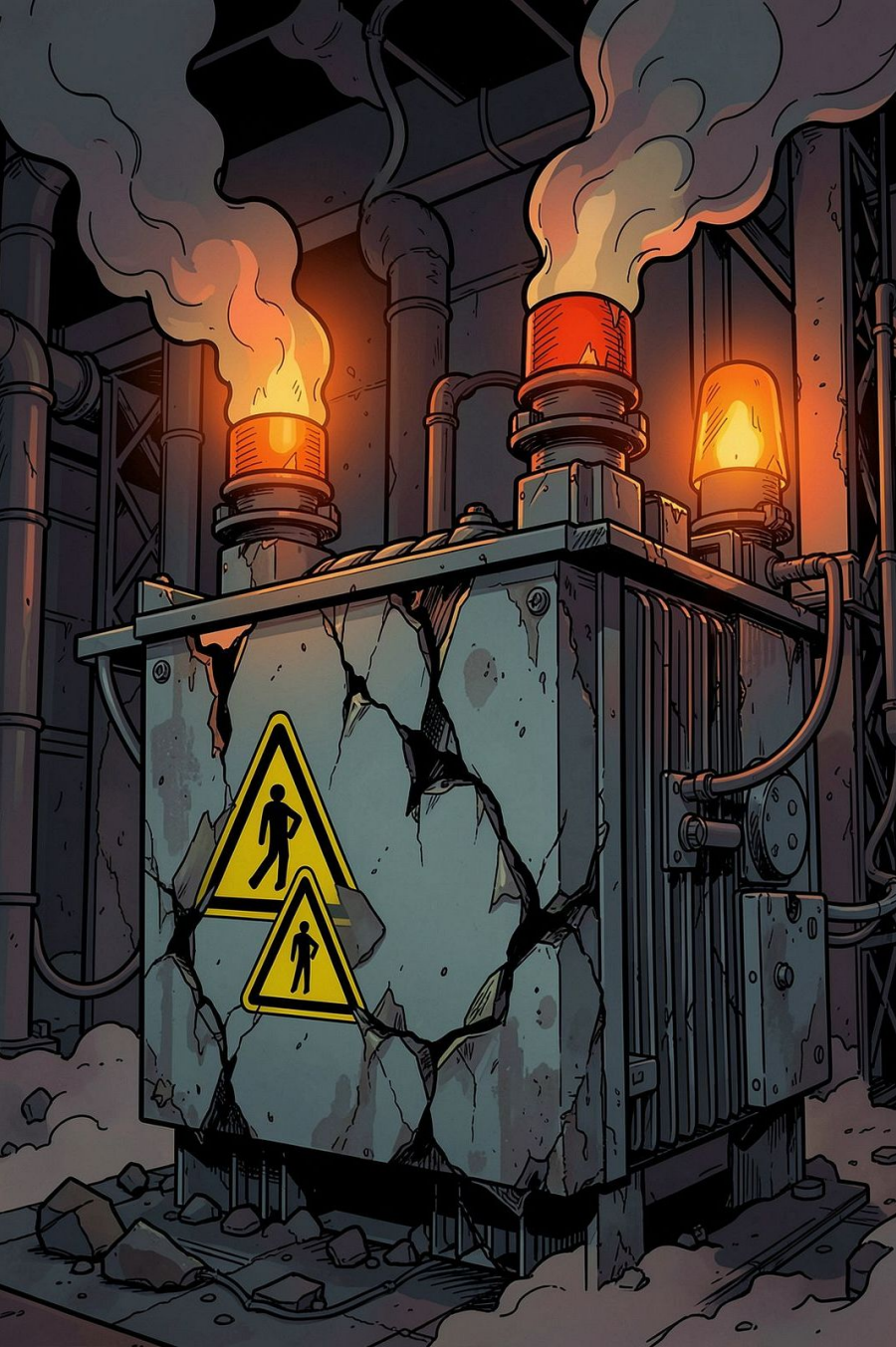




GridPulse AI

Transforming Power Grid Maintenance from Reactive to Predictive

Team: Achievers (AI Track)



The Problem

- Aging infrastructure plus unpredictable transformer failures lead to cascading blackouts and multi-million dollar damage.
- Current maintenance is reactive or calendar-driven — inefficient and costly.
- Field dispatch is chaotic: crews are sent without data-driven prioritization, wasting time and risking outages.



The Solution: GridPulse AI

GridPulse AI is an end-to-end predictive maintenance and intelligent dispatch system that: continuously ingests asset telemetry (including raw DGA), detects anomalies in real time, predicts failures before they occur, and automatically prioritizes high-risk assets for field crews.

- ❏ Reactive → Predictive: keep critical assets online by acting days or weeks earlier.

Core AI Capabilities



Fault Detection & Diagnosis

Multi-gas classification identifies failure modes such as Partial Discharge, Thermal Faults, and Arcing with per-gas attribution.



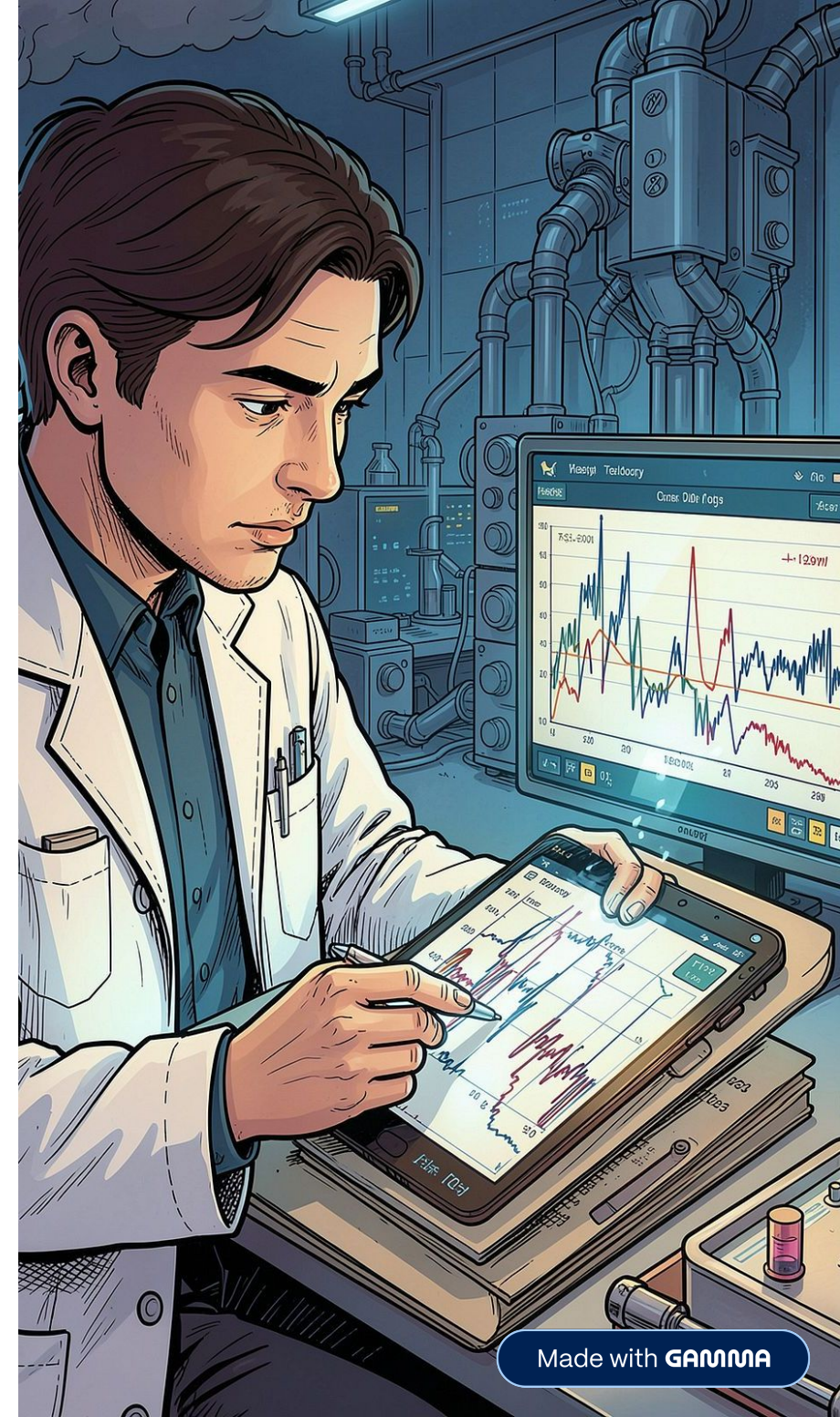
Remaining Useful Life (RUL)

Continuous cycle forecasting produces time-to-failure curves and confidence bands for operational planning.



Explainable AI (XAI)

Feature-level attributions highlight top physical risk factors — gas ratios, temperature trends, and load cycles — so engineers know why a score is high.



Architecture & Tech Stack

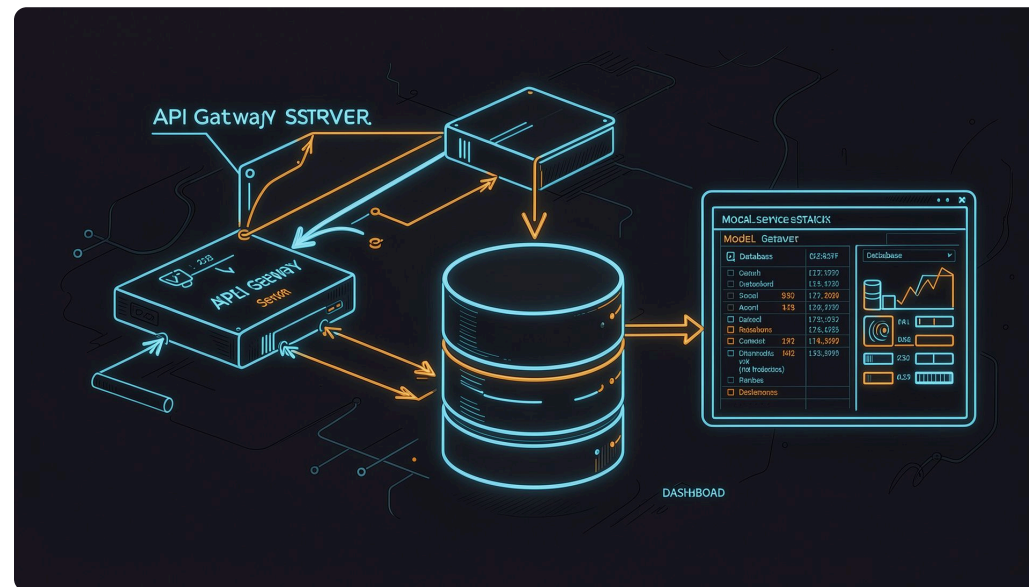
Frontend: React.js · Tailwind CSS · Lucide Icons — responsive realtime dashboard

Backend: Python · FastAPI — high-performance microservices, async endpoints

ML: Scikit-Learn · HistGradientBoosting · Isolation Forests — hybrid ensemble for detection + RUL

Database: Neon Serverless PostgreSQL — cloud-native time-series storage & transactional work logs

- Streaming: lightweight ingestion for raw DGA telemetry
- Model serving: low-latency endpoints for live inference
- Audit & immutability: append-only work history for regulatory compliance



The Product Experience



Live Dashboard

At-a-glance totals for assets, critical anomalies, and crew availability — real-time updates and alert prioritization.



Dynamic Filtering

Filter by geography, substation, asset type, or risk score to focus investigations and reduce noise.



Intelligent Dispatch

One-click "Pre-position Crew" assigns teams based on risk, ETA, and crew capacity; dispatch is logged immutably.



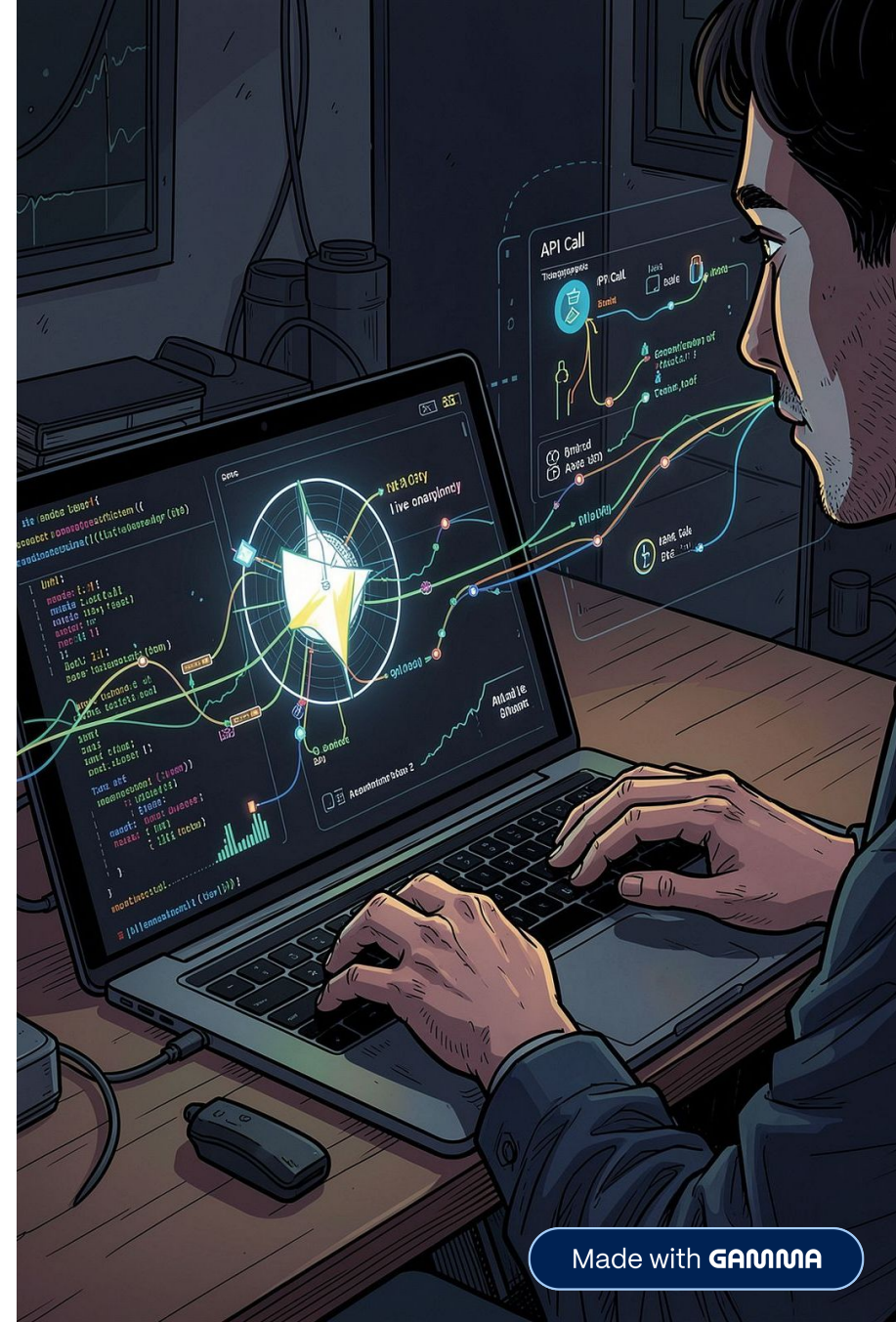
Immutable Work History

Audit-ready records show exact dispatch times, actions taken, and maintenance outcomes for compliance and review.

Live ML Inference in Action

- Closed-loop workflow: frontend modal accepts raw sensor readings ("Add Transformer"), posts to FastAPI endpoint.
- Backend synthesizes historical telemetry, runs ensemble inference (detection + RUL), and returns anomaly metadata and attribution.
- The dashboard renders the new anomaly instantly, updates prioritization, and enables immediate dispatch actions — no batch delays.

□ Low-latency inference (<500ms median) ensures field teams see emerging faults in near real time.





Business Impact & Value

Prevent Blackouts

Early detection prevents catastrophic failures, reducing outage frequency and duration.

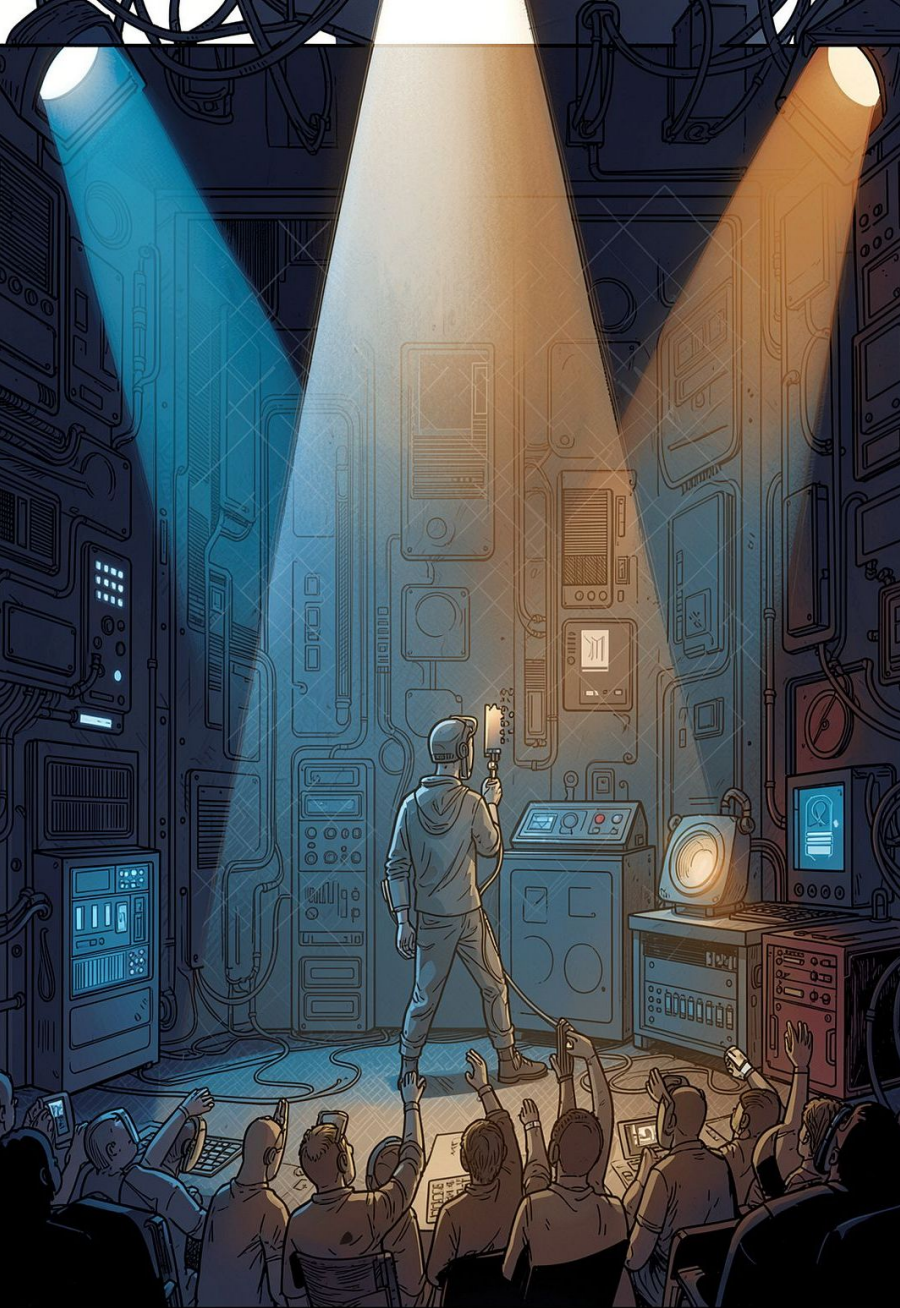
Cost Savings

Significant reduction in emergency replacements and outage penalties — millions saved annually for large utilities.

Optimized Workforce

Send crews only where math justifies action — lower travel costs, higher first-time fix rates, and improved safety.

Measurable KPIs: reduced unplanned failures, lower mean time to repair (MTTR), and improved asset utilization.



Thank You — Q&A

Team Achievers · Ready for live demo

Contact & Demo: request live walk-through of GridPulse AI and an example transformer inference pipeline.